

Date: September 15, 2026

To:

The Editor-in-Chief

Production (Associação Brasileira de Engenharia de Produção – ABEPRO / SciELO)

Subject: Submission of Original Research Article to *Production*

Dear Editor-in-Chief and Editorial Board,

We are pleased to submit our original research article entitled:

**“Online Constrained Truck Dispatching and Digital Modeling for Agro-Industrial
Receiving Yards: A Logistics 5.0 Perspective”**

for consideration for publication in *Production*.

Relevance and Fit to *Production*:

During seasonal harvest peaks, agro-industrial grain terminals experience severe bottlenecking due to physical contention for shared infrastructure—most notably shared weighbridges used for both gross and tare weighing. While traditional literature often approaches yard management via rigid First-In, First-Out (FIFO) rules or complex offline mathematical programs, these methods fail to adapt when real-world disruptions strike (e.g., rainfall over open hoppers, dumper mechanical failures, or administrative document holds).

Our study addresses this critical gap through the lens of **Logistics 5.0 in Production Engineering**. By formulating the terminal as a Dynamic Reentrant Hybrid Flow Shop (RHFS), we design and evaluate an online constrained dispatching decision-support engine that combines strict feasibility filtering with local short-horizon lexicographic ranking and an auditable event ledger. Our engine decouples reentrant contention, yielding a 44.8% reduction in peak queue waiting times (p95) under heavy congestion while fully preserving terminal throughput and guaranteeing zero operational constraint violations. Crucially, the system preserves human-centricity by providing explainable recommendations and supporting auditable supervisory interventions.

Statements of Originality and Ethics:

- This manuscript represents original work and has not been published previously.
- It is not currently under consideration for publication elsewhere.
- All authors have reviewed and approved the manuscript and agree with its submission to *Production*.
- The authors declare that there are no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

- The manuscript adheres to the journal's double-blind peer review requirements, with all author identifying information removed from the main text.

Thank you very much for your time and consideration of our manuscript. We look forward to receiving the reviewers' feedback.

Sincerely,

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